Group 7

Predictive Analytics for CitiBikeNYC

1. Problem Statement:

We are using CitiBikeNYC and weather data to build an integrated analytics dashboard and machine learning platform that explains past demand patterns, quantifies weather effects, predicts future ridership scenarios, and supports operational planning.

2. Dataset Link:

CitiBike Data: https://citibikenyc.com/system-data

Weather Data: https://www.ncei.noaa.gov/data/global-summary-of-the-day/archive/

3. Objective:

- To collect, clean, and integrate multi-year CitiBike trip data with corresponding historical weather data and handle missing values, outliers, and data inconsistencies to create a reliable analytical dataset.
- To analyze historical ridership trends across time (hourly, daily, monthly) and identify spatial demand patterns at station or neighborhood level and visualize peak usage periods.
- To quantify the effect of weather variables (temperature, rainfall, wind speed, extreme conditions) on bike ridership.
- To build machine learning models to forecast future CitiBike demand at station-level and city-level.
- To identify stations at risk of over-demand or under-utilization and to provide actionable insights for bike redistribution, maintenance planning, and capacity management.
- To develop an interactive analytics dashboard that combines historical analysis, weather insights, and demand forecasts.

4. **Data Dictionary:**

Sr. no	Column name	Column description	Datatype	Sample values
1	ride_id	A unique id assigned to each bike	String	BF31E940F7D80958, 0DF0EDCFF6452D83, 9AFD3BA9E53733E7, 146A0E14AC873008, 54D1DCA4B21AC7C1
2	rideble_type	Defines the type of the bike	String	electric_bike, classic_bike
3	started_at	Start time of the ride	Datetime	
4	ended_at	End time of the ride	Datetime	
5	start_station_name	Name of the station from wherevthe ride started	String	N 6 St & Bedford Ave, Huron St & Franklin St, Macon St & Howard Ave, Clark St & Henry St, Rivington St & Ridge St
6	start_station_id	A unique id assigned to each station	string	5379.1, 5869.04, 4408.07, 4789.03, 5406.02
7	end_station_name	Name of the station at which the ride ended	String	Broadway & Berry St, Stockholm St & Wilson Ave, Marcus Garvey Blvd & Macon St, E 51 St & 2 Ave, Union Ave & Wallabout St
8	end_station_id	A unique id assigned to each station	String	5164.05, 4824.03, 4278.03, 6575.03, 4818.03
9	start_lat	Latitude at which the ride began	Float	40.71745169, 40.73266, 40.68452, 40.69760127, 40.71850211
10	start_lgn	Longitude at which the ride began	Float	-73.95850939, -73.95826, -73.92011,

11	end_lat	Latitude at which the ride finished	Float	40.7103612992258, 40.699304, 40.682601, 40.755293, 40.70029511
12	end_lgn	Longitude at which the ride finished	Float	-73.9653038978576, -73.923044, -73.938037, -73.967641, -73.95032283
13	member_casual	Tells whether the user using the bike is a casual member (who uses the bike rarely) or a frequent member(who uses bike regularly)	String	Member, casual
14	trip_date	Date at which the ride took place	Date	11-07-2024, 12-07-2024
15	Station	Station number of weather office	String	72505394728
16	Latitude	Given in decimated degrees (Southern Hemisphere values are negative)	Float	40.77898
17	Longitude	Given in decimated degrees (Western Hemisphere values are negative)	Float	-73.96925
18	Elevation	Elevation of the station above sea level in meters.	Float	42.7
19	Name	Name of station/airport/military base.	string	NY CITY CENTRAL PARK, NY US
20	Temp	Mean temperature of the day (°F × 0.1 precision).	float	76.2
21	Temp_Attributes	Number of hourly observations used to compute the mean temperature.	int	24
22	Dewp	Mean dew point temperature (°F × 0.1).	float	68.2

23	Dewp_Attributes	Number of observations used to compute dew point.	int	24
24	SLP	Mean sea level pressure (millibars × 0.1).	float	1016.1
25	SLP_Attributes	Number of observations used for sea level pressure.	int	20
26	STP	Mean station pressure (millibars × 0.1)	float	11.3
27	STP_Attributes	Number of observations used for station pressure.	int	24
28	VISIB	Mean visibility in miles (× 0.1).	float	9.3
29	VISIB_Attributes	Number of observations used to compute visibility.	int	24
30	WDSP	ean wind speed (knots × 0.1).	float	2.2
31	WDSP_Attributes	Number of observations used in wind speed calculation.	int	22
32	MXSPD	Maximum sustained wind speed (knots × 0.1).	float	5.1
33	Gust	Maximum wind gust for the day (knots × 0.1).	float	999.9
34	Max	Maximum temperature recorded (°F × 0.1).	float	89.1
35	Max_Attributes			
36	Min	Minimum temperature recorded (°F × 0.1).	float	71.1
37	Min_Attributes			
38	PRCP	Total precipitation (inches × 0.01).	float	0.18
39	PRCP_Attributes	**Meaning:** Indicates source type: • A → 1 report of 6-hour precipitation • B → Sum of 2 × 6-hour reports • C → Sum of 3 × 6-hour reports	String	G

		• D → Sum of 4 × 6-hour reports • E → 12-hour report • F → Sum of 2 × 12-hour reports • G → 24-hour report • H → Station reported 0 but precipitation occurrence recorded • I → No precipitation reported		
40	SNDP	Snow depth in inches × 0.1.	float	999.9
41	FRSHTT	**Meaning:** Weather event indicators (6 digits): • Digit 1 → Fog • Digit 2 → Rain/Drizzle • Digit 3 → Snow/Ice Pellets • Digit 4 → Hail • Digit 5 → Thunder • Digit 6 → Tornado / Funnel cloud • **1 = event occurred**, **0 = not reported**	int	0

5. Key Performance Indicators:

User Segmentation:

1. Member vs casual usage patterns
2. Weather sensitivity index by user type
3. Preferred bike type by weather condition

Time based KPI's:

1. Average Trip duration(end_time - start_time)
2. Peak riding hours
3. Weekend vs weekday usage
4. Trips by time of the day

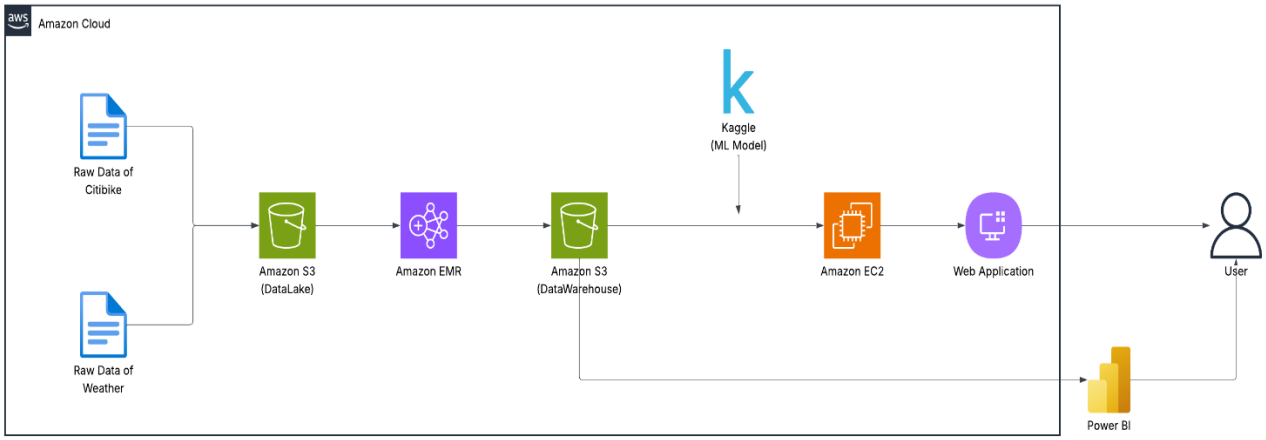
Temperature Impact:

1. Average rides on hot and cold days.
2. Average rides on rainy days vs clear days.
3. Impact of Visibility.
4. Optimal temperature range for maximum ridership

Trip Distance & Geographic KPIs:

1. Average Trip Distance (km)
2. Longest/Shortest Trips

Station Performance KPIs:

1. Top 10 Start Stations
2. Station Demand Ranking (monthly): lists the most used start/end stations during the month.
3. Station Imbalance Index(monthly): tracks inflow vs outflow difference per station each month.

6. Architecture Diagram: